

The application debugging tool

Let's discuss Android Debug Bridge in this section.

Android Debug Bridge

This is also called ADB. It is a software interface for the Android development system. ADB is part of the Android SDK (Android Software Development Kit) and is placed in the sub-directory called *platform-tools*.

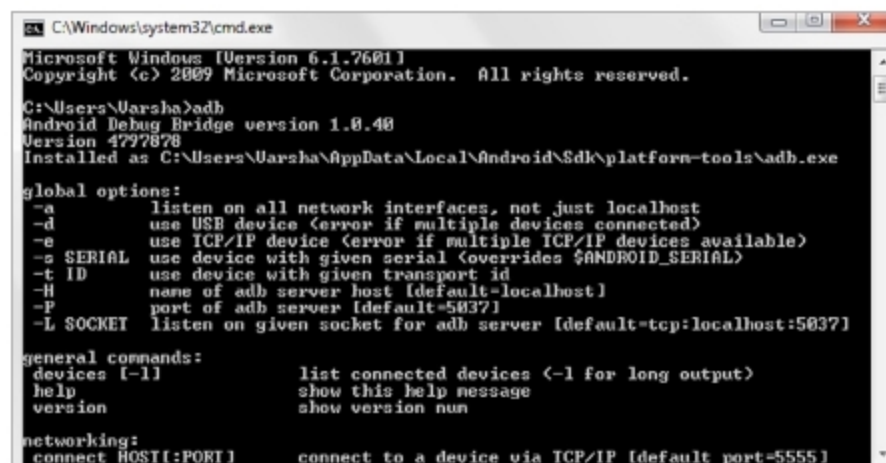
Setting up ADB on Windows: When you are planning to test and debug your .apk file in real devices, then some setting up is required on your phone or system.

For the phone setup:

1. Go to your phone *Settings* application.
2. Tap the *About Phone* option.
3. Tap the *Build number* option seven times to enable the *Developer Mode*. You will see the toast message on your phone once the *Developer Mode* is on.
4. Then go back to the main *Settings* screen and select the *Developer Mode* option.
5. Next, enable the *USB Debugging mode* option.

For installing the SDK manager on Windows:

1. Install the SDK manager from Android Studio or else download the SDK platform tools for Windows from the Android-Developer website.
2. Set the proper path where your SDK bundle is available in your system.



```

C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
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C:\Users\Varsha>adb
Android Debug Bridge version 1.0.40
Version 4777878
Installed as C:\Users\Varsha\AppData\Local\Android\Sdk\platform-tools\adb.exe

global options:
-a          listen on all network interfaces, not just localhost
-d          use USB device (error if multiple devices connected)
-e          use TCP/IP device (error if multiple TCP/IP devices available)
-s SERIAL   use device with given serial (overrides $ANDROID_SERIAL)
-t ID       use device with given transport id
-H          name of adb server host (default=localhost)
-P          port of adb server (default=5037)
-L SOCKET   listen on given socket for adb server (default=tcp:localhost:5037)

general commands:
devices [-l]      list connected devices (-l for long output)
help             show this help message
version          show version num

networking:
connect HOST[:PORT]  connect to a device via TCP/IP [default port=5555]
  
```

Figure 5: *adb* command

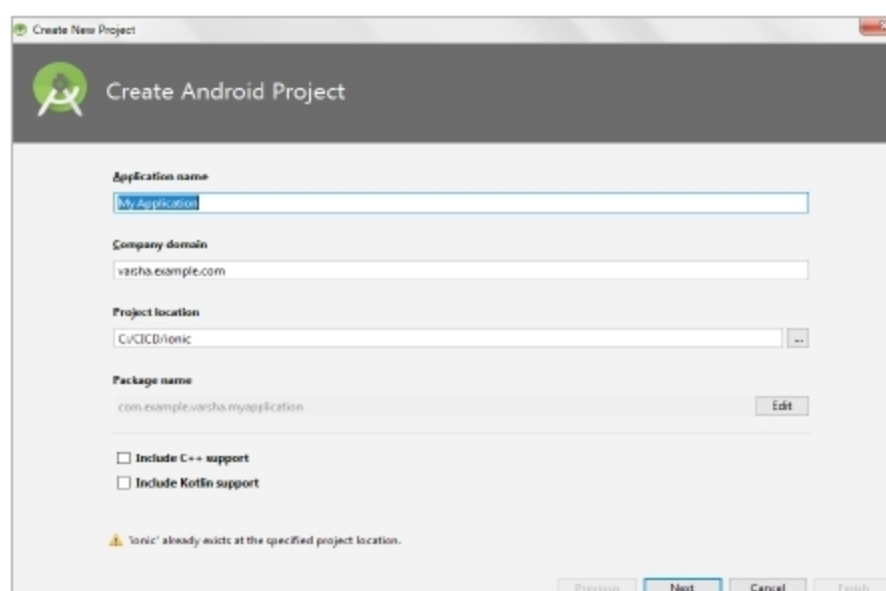


Figure 6: Android Project

3. Now set these paths for the *Detect ADB* path in your environment variable, as follows:

```

C:\Users\Varsha\AppData\Local\Android\Sdk\build-tools;
C:\Users\Varsha\AppData\Local\Android\Sdk\platform-tools;
C:\Users\Varsha\AppData\Local\Android\Sdk\tools;
C:\Users\Varsha\AppData\Local\Android\Sdk\platforms
  
```

4. Now go to command prompt and check ADB.
5. Connect the device via USB and test whether ADB devices are working or not by using the following command:

```
adb devices
```

6. Now you can install the .apk file in your real device through this command. Before running the command, go to the proper path where your .apk file is present, then run the following command:

```
adb install "your apk file name"
```

This command will install the .apk file in your device.

Stable release	Android-28
Download link	From the <i>Android-Developer</i> website
Operating systems	Windows, MacOS, Linux
Main features	Executes the command on the phone directly. Debugging of code.

IDE

Android Studio

Android Studio is an open source and Linux based operating system that is used for mobile device development. Android Studio works based on IntelliJ IDEA. This is used for developing native and hybrid applications.

Requirements for setting up Android Studio:

1. Java JDK5 or latest version
2. Java Runtime Environment (latest version)
3. Android Studio

Installation in Windows:

1. Go to the Android Studio official website and download the latest version.
2. Install Android Studio in your Windows system.
3. Once Android Studio is open, start a new project.
4. Write your project's name and click the *Next* button.
5. Set the Android target and click *Next*.
6. Select *Empty Project*.
7. Set the activity name.